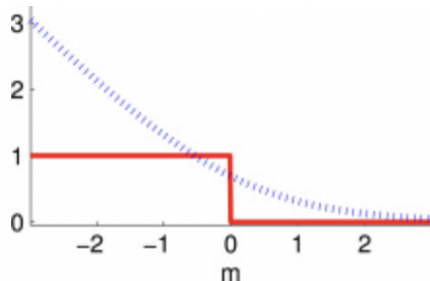


How to Choose K?

Procheta Sen



Effect of K

The Parameter K .

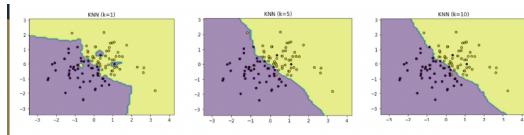


Figure: Effect of Choosing Different Values of K

The parameter k

- ▶ k is a *hyperparameter* of the algorithm.
- ▶ What value to use for k ?
 - ▶ Depends on the dataset size. Large and diverse datasets need a higher k , whereas a high k for small datasets might cross out of the class boundaries.
- ▶ How to choose k ?
 - ▶ **A bad option:** try different values of k when evaluating on **test data**.
 - ▶ Use **validation dataset** to find a good value of k .

Train / Validation / Test approach

Validation data

- ▶ Set aside (hold-out) a fraction of train data for validation purposes.
- ▶ Using validation data to set hyperparameters reduces overfitting.
- ▶ Of course, you **CANNOT** use test data for any tuning except for testing.



Figure: Train/Test Split

Example of Cross-validation



Figure: K fold Validation

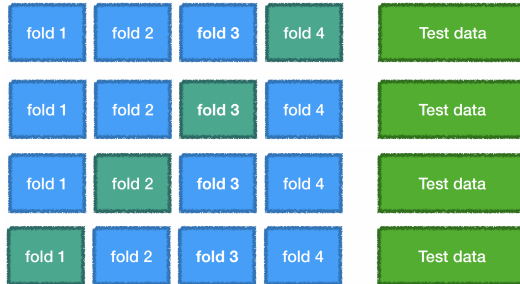


Figure: Demonstration of K fold Validation